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Constituent of Symbiosis International (Deemed University), Pune

Department of Artificial Intelligence and Machine Learning

Lab: Data Visualization Lab

Faculty: Dr. Shivali Wagle

Aim: To develop a single-page sales analysis dashboard applying data visualization best practices

Datasets Used:

1. AdventureWorks Sales Data 2022
2. AdventureWorks Calendar Lookup
3. AdventureWorks Customer Lookup
4. AdventureWorks Product Categories Lookup
5. AdventureWorks Product Lookup
6. AdventureWorks Product Subcategories Lookup
7. AdventureWorks Territory Lookup

Additional tables created during Power Query exercises:

8. Student
9. JanuarySales
10. FebruarySales
11. ProductID

These additional tables were created using **Enter Data** specifically for demonstrating Pivot, Unpivot, Append and Merge operations.

Objectives and Implementation:

The practical contains two parts:

1. Power Query operations
2. Sales Analysis dashboard

The practical specifically includes Grouping, Pivoting/Unpivoting, Append and Merge operations before the dashboard.

I. Importing the Dataset

The following queries were loaded:

- Sales Data 2022
- Calendar Lookup

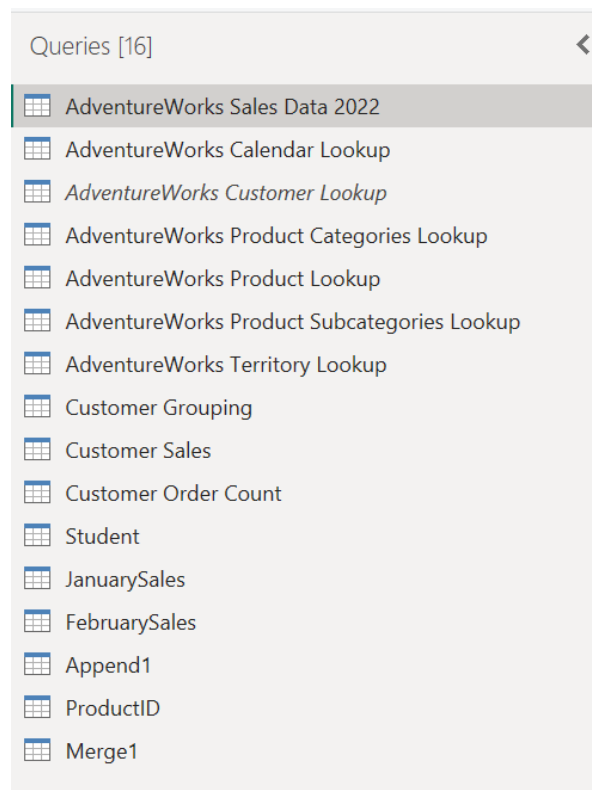


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- Customer Lookup
- Product Categories Lookup
- Product Lookup
- Product Subcategories Lookup
- Territory Lookup



II. Grouping

The practical asks 3 questions:

1. How many distinct customers are there?
2. What is the average sales per customer?
3. Which customer has placed the most orders?

For all three I used basic Group By, because each operation required only one field/column and one aggregation



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1. Number of Distinct Customers

- Right-clicked **AdventureWorks Sales Data 2022**.
- Selected **Reference**.
- Created a new query named **Customer Grouping**.
- Selected **Home** → **Group By**.
- Used **Basic Group By**.
- Selected:
 - **Group by:** CustomerKey
 - **New column name:** Order Count
 - **Operation:** Count Rows
- Clicked **OK**.
- Each row represented a unique customer.
- The number of rows in the resulting grouped table represented the number of distinct customers.

CustomerKey	Order Count
23791	3
16747	3
11792	2
11530	9
18155	3
13541	2
18259	1
23694	3
22308	4
20236	1
26273	3
19709	6
21239	2
17252	3
17396	2
27148	2
20260	2
12859	5
20454	2
14751	2
27477	2
23011	2
11891	1
13625	5
14486	3
15050	3
12411	3
12627	5
23431	3
13996	1
11306	3



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2. Average Sales per Customer

Since the AdventureWorks Sales table did not directly contain a Sales column, sales had to be derived using product information.

- Created a **Reference** of AdventureWorks Sales Data 2022.
- Renamed the query to **Customer Sales**.
- Used **Home** → **Merge Queries**.
- Merged Sales Data 2022 with **Product Lookup**.
- Used ProductKey as the common column.
- Used a **Left Outer** join.
- Expanded the merged table.
- Selected only ProductPrice.
- Created a Custom Column named **Sales**.
- Used the formula: [OrderQuantity] * [ProductPrice]
- Changed the Sales column to **Decimal Number**.
- Used **Group By** → **Basic**.
- Grouped by CustomerKey.
- Selected:
 - **New column:** Average Sales
 - **Operation:** Average
 - **Column:** Sales

The sales value was calculated at the order-line level by multiplying the quantity ordered by the product price. Grouping these values by customer allowed the average sales value to be calculated for each customer.



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Assignment4

File

Home

Transform

Add Column

View

Tools

Help

Close & Apply

New Source

Recent Sources

Enter Data

Data source settings

Data Sources

Parameters

Export query results

Output Data

Refresh Preview

Advanced Editor

Query Management

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Reduce Rows

Sort

Split Columns

Group By

Data Type: Date

Use First Row as Headers

Replace Values

Transform

Merge Queries

Append Queries

Combine Files

Combine

Queries [16]

</

Assignment4

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Query Choose Remove Keep Split Group Data Type: Whole Number Merge Queries Append Queries Combine Files Combine

Queries [16]

AdventureWorks S... AdventureWorks C... AdventureWorks C... AdventureWorks P... AdventureWorks P... AdventureWorks T... Customer Grouping Customer Sales Customer Order C... Student JanuarySales FebruarySales Append1 ProductID Merge1

Table.Group(*"changed Type", ("CustomerKey"), [{"Average Sales", each List.Average([Sales], type nullable number)}])

1.1	1.2
CustomerKey	Average Sales
23791	25.19
16747	740.7141667
26273	28.2014
15146	28.81376667
20262	19.84333333
24394	20.29
11262	22.4047975
16436	564.944175
20974	45.51666667
24585	273.010425
18155	17.8714
11891	34.99
12473	319.45
11464	1209.53
11201	33.94025
20683	867.99
28943	21.65
20135	26.65
27065	43.98
16717	36.16434286
11280	19.3175
15684	700.4588667
13050	833.48455
17184	657.2905429
11792	40.85575
23694	34.53473333
20260	867.3171
12563	43.417875
14581	190.03608
16720	21.8121
18492	60.29605

2 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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3. Customer with the Most Orders

The practical specifies grouping by customer and counting orders.

- Created another **Reference** of AdventureWorks Sales Data 2022.
- Renamed it **Customer Order Count**.
- Selected **Home** → **Group By**.
- Used **Basic Group By**.
- Selected:
 - **Group by:** CustomerKey
 - **New column:** Order Count
 - **Operation:** Count Rows
- Clicked **OK**.
- Sorted Order Count in descending order.
- The customer at the top had the highest order count.

Query: = Table.Sort(*Grouped Rows*, [{"Order Count", Order.Descending}])

CustomerKey	Order Count
11300	45
11185	45
11262	40
11276	36
11223	33
11330	31
11091	30
11277	29
11711	28
11566	25
11200	23
11211	22
11287	22
11331	22
11176	21
11142	21
11500	20
11619	19
11501	18
11203	17
11505	16
11019	16
11502	16
11212	16
11631	14
11253	14
11507	14
11506	14
11429	14
12074	13
11420	13
11221	12
13096	12



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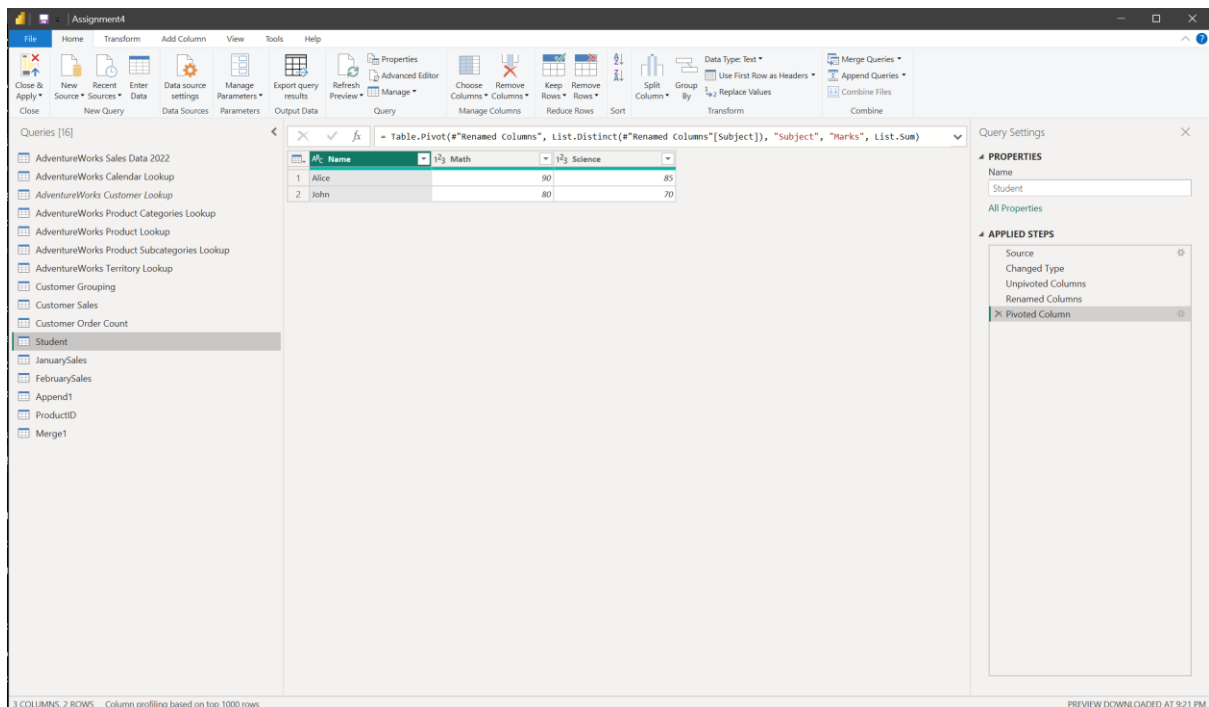
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III. Pivoting and Unpivoting

The practical asks us to create a Student table, unpivot Math and Science, rename the resulting columns, and then pivot the table back.

Created a Student table using Home-> Enter Data



1. Unpivoting

- Selected the **Math** and **Science** columns.
- Went to **Transform** → **Unpivot Columns**.
- The columns were converted into rows.
- The generated columns were:
 - Attribute
 - Value
- Renamed:
 - Attribute → Subject
 - Value → Marks



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Table.UnpivotOtherColumns(#\"Changed Type\", {\"Name\", \"Attribute\", \"Value\"})

	Name	Attribute	Value
1	John	Math	80
2	John	Science	70
3	Alice	Math	90
4	Alice	Science	85

3 COLUMNS, 4 ROWS Column profiling based on top 1000 rows

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2. Pivoting

- Selected the `Subject` column.
- Selected **Transform** → **Pivot Column**.
- Used `Subject` as the column to pivot.
- Selected `Marks` as the **Values Column**.
- Applied the transformation.



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3 COLUMNS, 2 ROWS Column profiling based on top 1000 rows

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IV. Append Queries

The practical demonstrates Append using JanuarySales and FebruarySales.

JanuarySales and FebruarySales were created.

2 COLUMNS, 2 ROWS Column profiling based on top 1000 rows

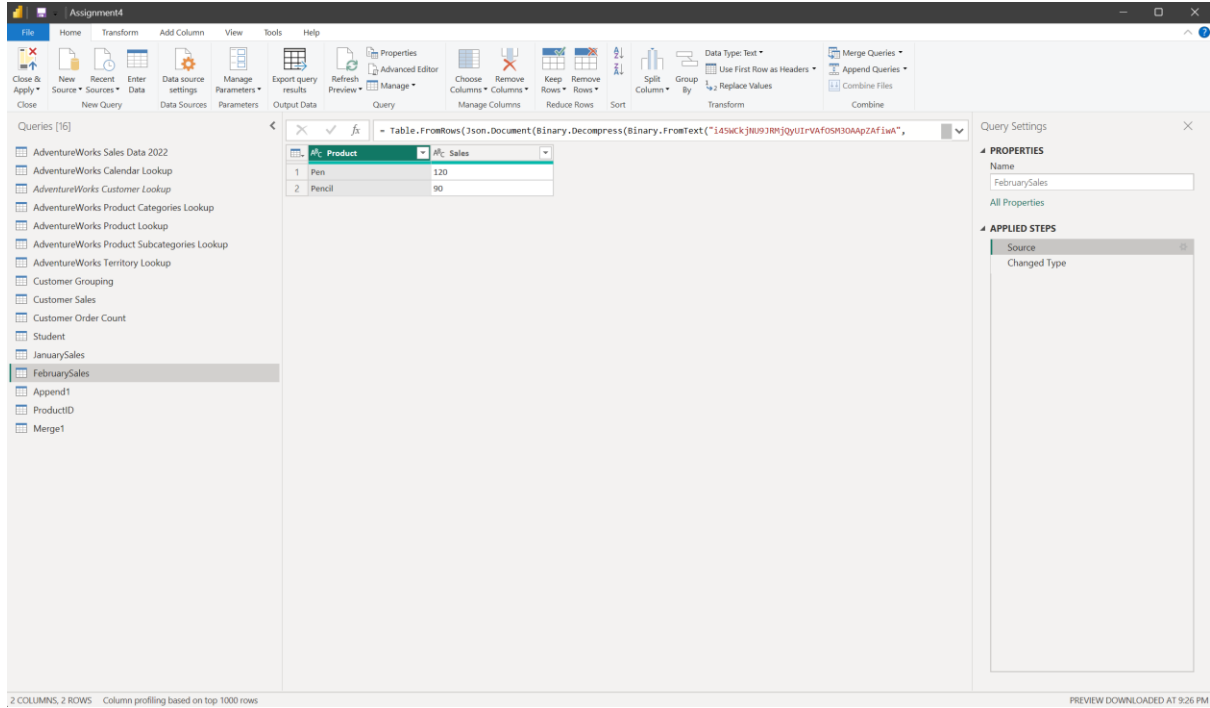
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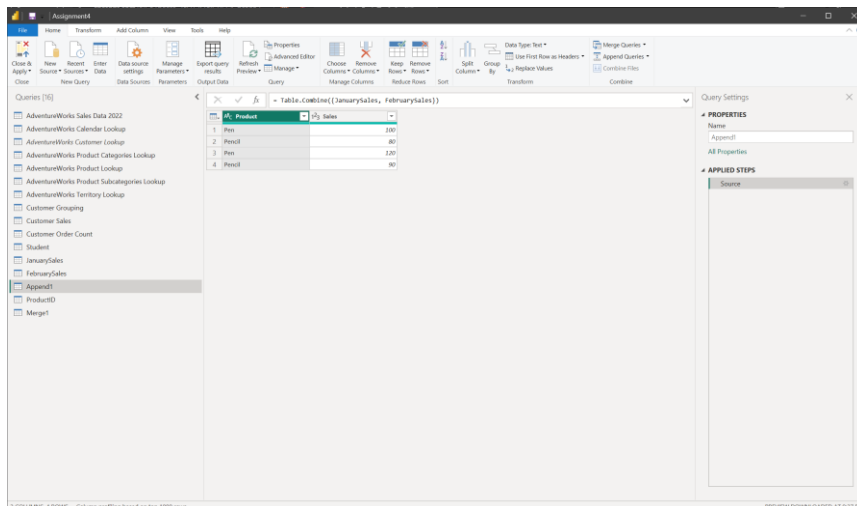
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- Opened Power Query Editor.
- Selected:
Home → Append Queries → Append Queries as New
- Selected:
 - First table: JanuarySales
 - Second table: FebruarySales
- Clicked **OK**.
- The resulting query was named **Append1**.





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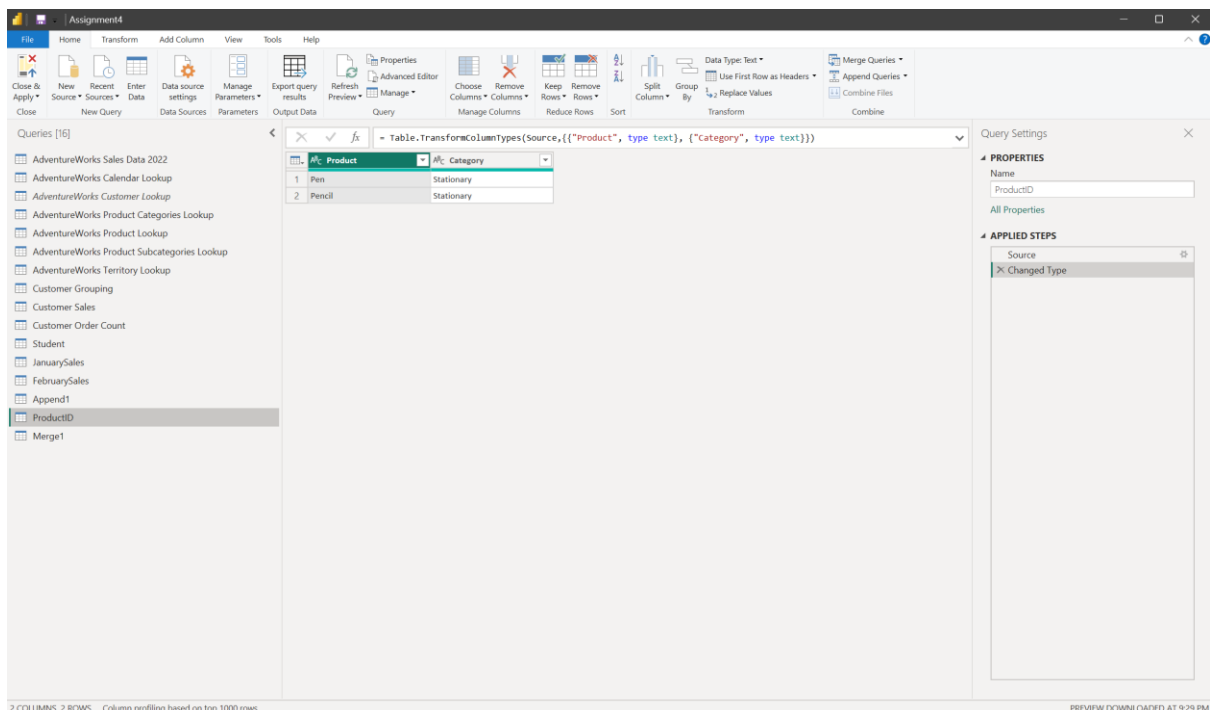
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Append is used to **combine rows from two tables vertically** when the tables have the same or compatible column structure.

V. Merge Queries

The practical asks for a ProductID table where Product is the common column with JanuarySales, followed by selecting a join kind and expanding the merged table.

- Created a new table using **Enter Data**.
- Named it **ProductID**.



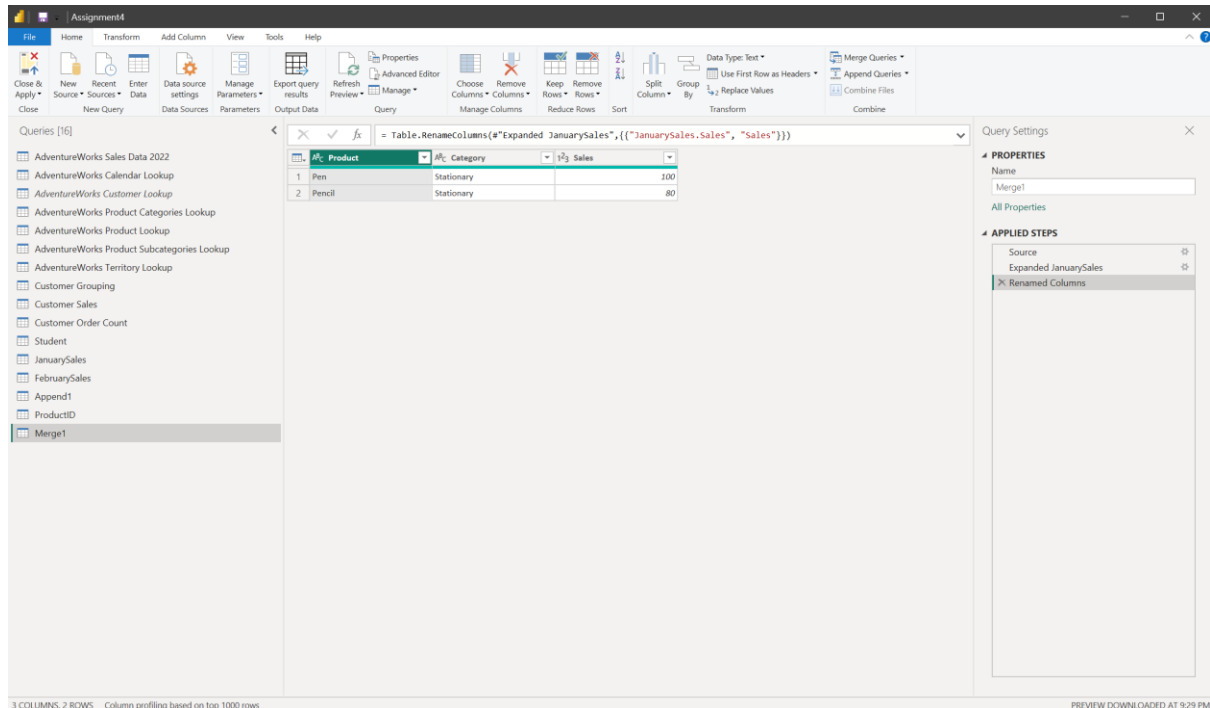
- Selected ProductID.
- Selected **Home** → **Merge Queries** as **New**.
- Selected JanuarySales as the second table.
- Selected Product as the common column in both tables.
- Used a **Left Outer** join.
- Clicked **OK**.
- Expanded the merged table.
- Selected the Sales column.



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Merge is used to **combine columns from related tables using a common key/field**. Unlike Append, which stacks rows, Merge combines information horizontally.

VI. Sales Analysis Dashboard

After completing the Power Query exercises, the transformed data was loaded into Power BI and a single-page Sales Analysis Dashboard was created.

The practical requires the dashboard to contain four KPI cards, four analytical visuals, a customer-wise table and three slicers.

1. KPI Cards

Four Card visuals were created:

- **Total Sales** – displays overall sales.
- **Total Profit** – displays overall profit.
- **Total Orders** – displays the total number of orders.
- **Total Customers** – displays the number of unique customers.

2. Sales by Category

A **Clustered Column Chart** was created using CategoryName and Total Sales to compare sales across product categories.



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3. Monthly Sales Trend

A **Line Chart** was created using Month Name and Total Sales to show monthly sales trends. Month Name and Month Number columns were created to display the months in chronological order.

4. Sales by Region

A **Donut Chart** was created using Region and Total Sales to show the contribution of different regions to overall sales.

5. Top 10 Product by Sales

A **Bar Chart** was created using ProductName and Total Sales. A **Top 10 filter** was applied to display the ten highest-selling products.

6. Customer-wise Sales

A **Table** was created using CustomerKey and Total Sales to display sales for individual customers.

7. Slicers

Three interactive slicers were added:

- **Order Date** – filters the dashboard by date.
- **Region** – filters sales by region.
- **Category** – filters sales by product category.





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Conclusion

The experiment successfully demonstrated the use of **Microsoft Power BI** for transforming sales data and creating an interactive Sales Analysis Dashboard. Power Query was used for grouping, pivoting, unpivoting, appending and merging data, while cards, charts, tables and slicers were used to analyze sales performance.

The final dashboard provides insights into **sales, profit, orders, customers, categories, regions, products and monthly trends**, while interactive slicers allow users to filter the information according to their requirements.